

TITLE PAGE

- Problem Statement ID – **SIH26103**
- Problem Statement Title– Use case on web-based integrated project-monitoring platform
- Theme– **Smart Automation**
- PS Category– **Software**
- Team ID– **SIH-68**
- Team Name (Registered on portal) – **Nirvana**



Vikas Drishti transforms PAIMANA from “monitoring what happened” to “anticipating what happens next.” It creates a connected infrastructure intelligence ecosystem from Ministers and policymakers to project teams, field officers and contractors predicting risks, explaining why they occur, recommending interventions and verifying action on the ground.

Innovation and Uniqueness

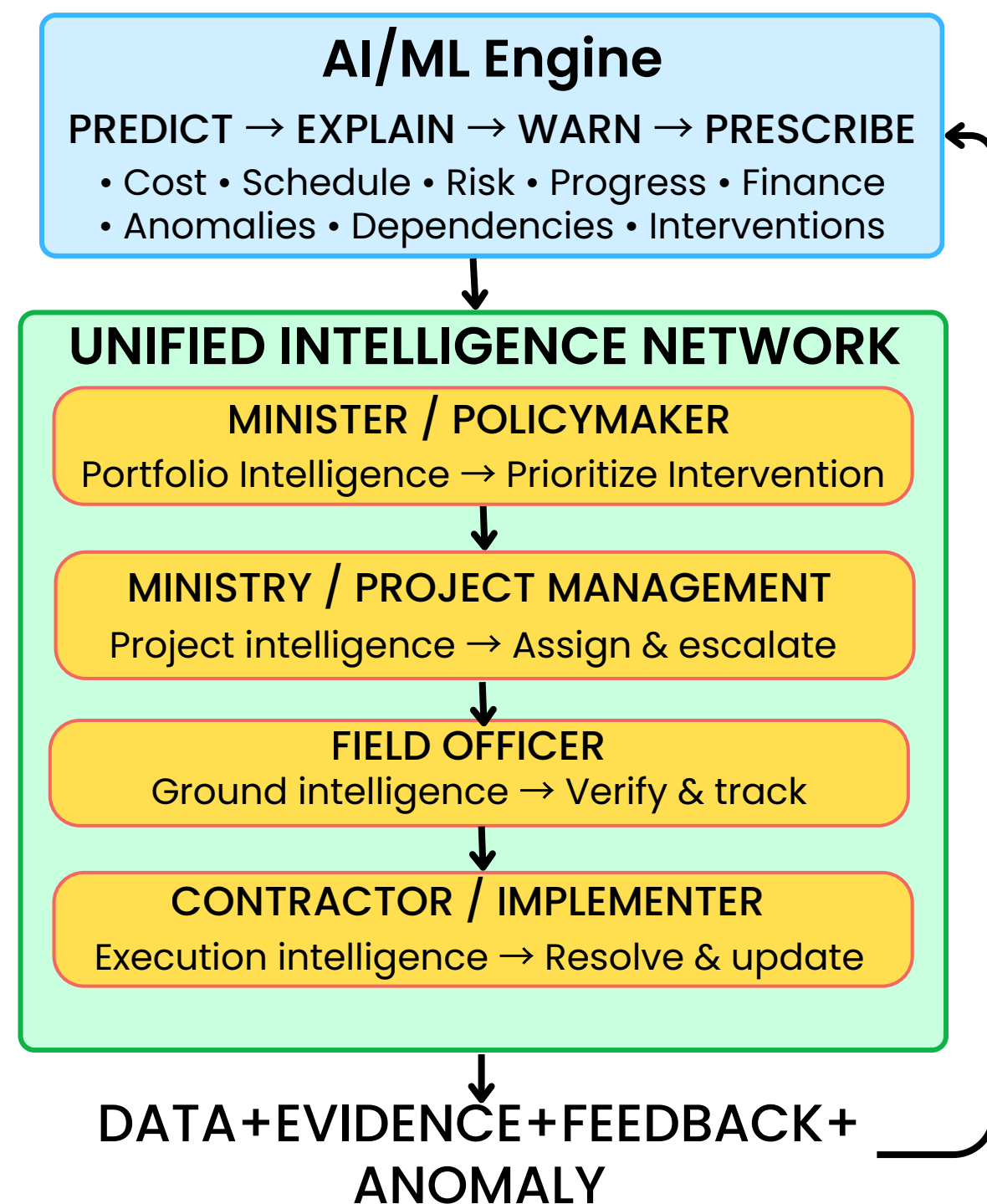
15–30 day early warnings for cost, schedule & implementation risks using dynamic scoring and sector-specific models for Roads, Power, Railways, etc.

Explainable intelligence with SHAP, RAG assistant, CUF-vs-enriched data analysis and an India-wide geospatial risk heatmap for decision support.

From risk to action: What-If policy simulation, ranked interventions and dependency graphs identify downstream impacts and the highest-priority corrective actions.

Integrity & verification: anomaly/red-flag detection with PFMS expenditure matching and satellite/Bhuvan progress verification, on sovereign 100% on-prem AI.

VikasDrishti Ecosystem



How It Addresses the Problem



Predict Before It's Too Late

Moves PAIMANA from descriptive → predictive → prescriptive, forecasting delays, cost overruns and implementation risks with explainable AI.



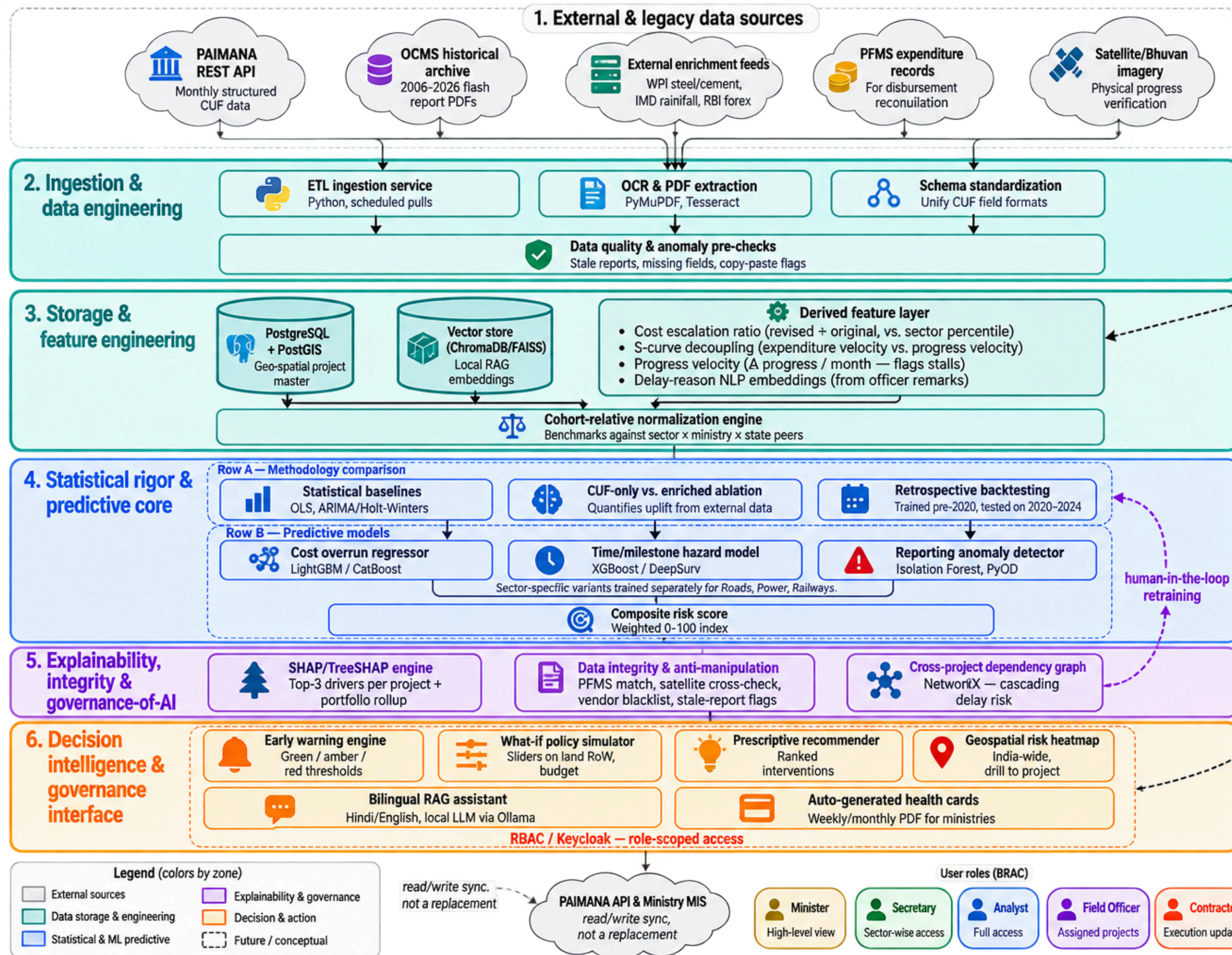
INTELLIGENCE ECOSYSTEM

Connects Ministers → Project Managers → Field Officers → Contractors, enabling intelligence from portfolio to ground execution.



MONITOR → LEARN → IMPROVE

Creates a feedback loop using field evidence, validation, anomaly detection and audit trails for smarter governance.

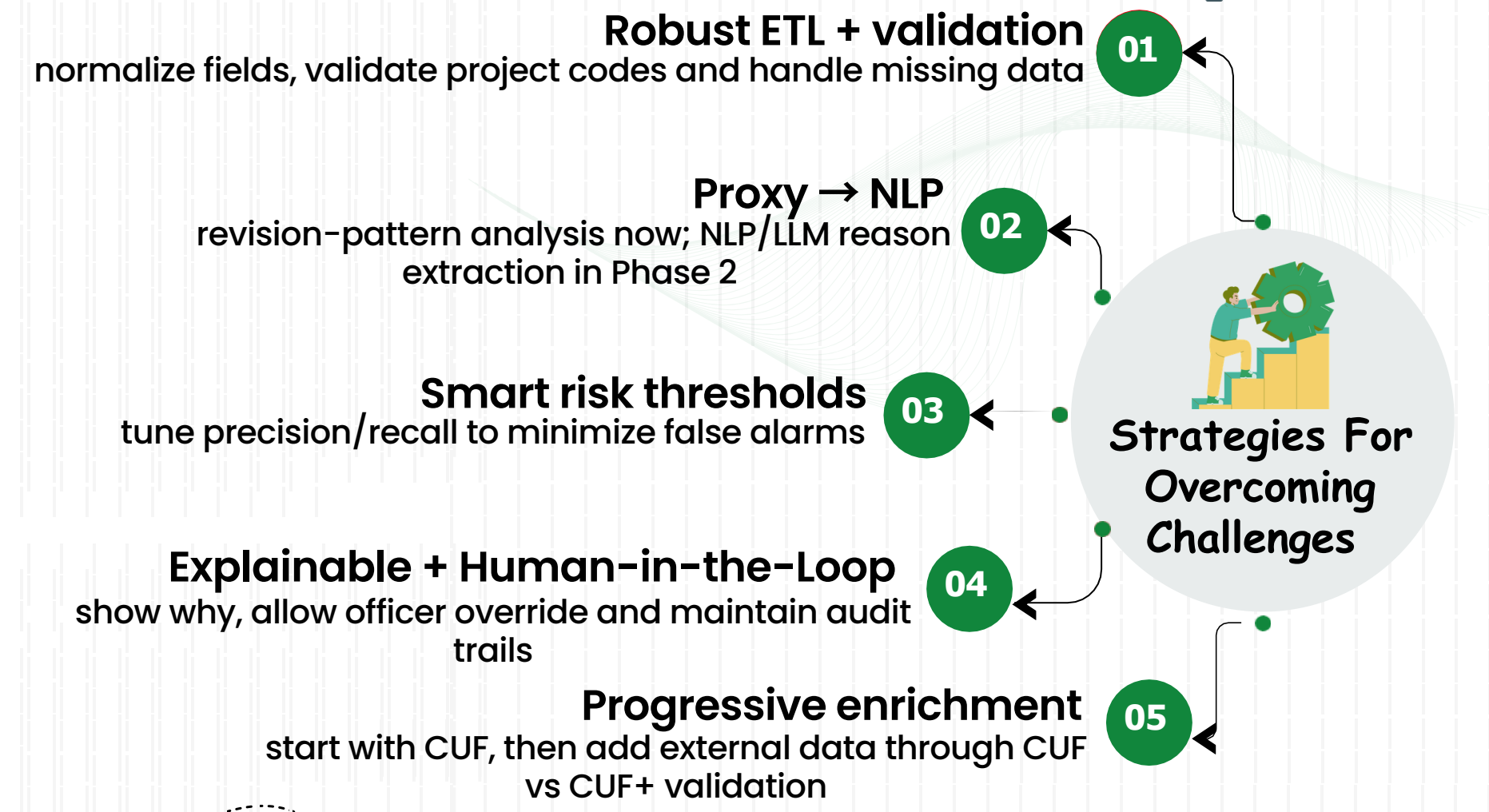
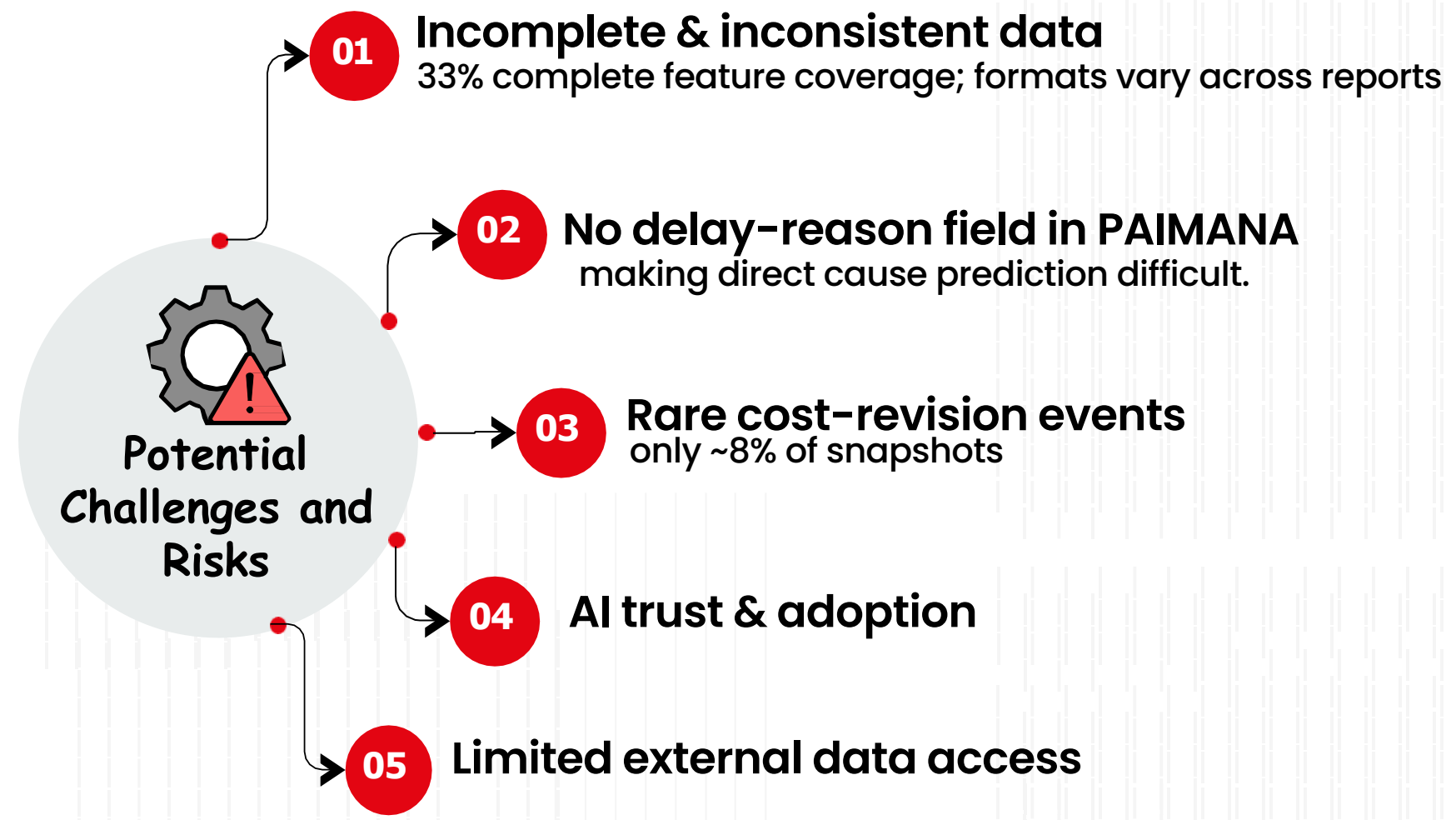


Answers PS requirement:
quantify AI/ML gain
vs. statistics

S-curve decoupling caught a real 34.7% cost jump in one June→July flash report diff

MVP-ready in phase 1;
PFMS/satellite links are
phase-2 conceptual
integrations





Feasibility

Proven on Real Government Data

Tested on PAIMANA Flash Reports – 18,124 rows, 2,304 projects, 14 monthly snapshots (Apr 2025–Jul 2026), with 20 years of OCMS history available for deeper training and validation.

Predictive Performance Validated

XGBoost achieves ROC-AUC 0.83–0.94 for next-month cost/schedule revisions using project-level, leakage-safe splits; benchmark against statistical models and CUF-only vs CUF+ enriched data.

Seamless PAIMANA Integration

Uses existing PAIMANA/CUF monthly/API feeds and derives additional features from existing fields—no parallel data-entry system or workflow change for field officers.

Sovereign & Scale-Ready

Open-source, no licensing cost, on-prem/NIC-cloud ready; external data enrichment, NLP/LLM-based reason extraction and historical OCMS integration can be added through a phased 2–3 month roadmap.

Potential Impact on the Target Audience

Ministers & Policymakers — SEE & DECIDE

- Portfolio-wide visibility
- Risk & bottleneck identification
- Resource prioritization
- Data-driven policy decisions

Project Managers — MONITOR & MANAGE

- Centralized project intelligence
- Real-time progress & milestones
- Early risk detection
- Faster, informed decisions

Field Officers & Contractors — ACT & EXECUTE

- Ground-level issue reporting
- Clear tasks & deadlines
- Resource coordination
- Corrective-action tracking

Stakeholders & Public — TRUST & BENEFIT

- Greater transparency
- Clear accountability
- Better fund/resource utilization
- Faster, more reliable infrastructure delivery

Benefits of Vikas Drishti

Economic Impact

- Control cost escalation & reduce avoidable delays
- Improve utilisation of public funds
- Accelerate infrastructure asset delivery

Social Impact

- Faster delivery of roads, railways, power, water & social infrastructure
- Earlier public access to essential services

Environmental Impact

- Identify stalled projects earlier
- Improve resource utilisation & reduce wastage
- Minimise unnecessary rework

Governance Impact

- Evidence-based decision-making
- Transparent & explainable AI
- Stronger auditability & accountability
- Better inter-agency coordination

Transparency & Integrity

- Detect financial & progress anomalies
- Flag unusual project patterns
- Create traceable audit trails
- Reduce opportunities for leakage & manipulation

Parliament Q&A link: <https://sansad.in/lis/questions/questions-and-answers>
link to access data: <https://paimana-proj.mospi.gov.in/ReportPage>

PAIMANA portal (official): <https://paimana-proj.mospi.gov.in>

PIB / official PAIMANA info: <https://informatics.nic.in/files/websites/october-2025/paimana-portal.php>

PM Gati Shakti National Master Plan: <https://www.digitalindia.gov.in/initiative/gati-shakti/>

PRAGATI platform: https://www.pmindia.gov.in/en/news_updates/pm-launches-pragati-a-multi-purpose-multi-modal-platform-for-pro-active-governance-and-timely-implementation/

PFMS (Public Financial Management System):

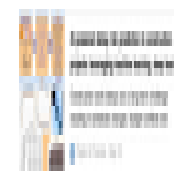
July 2026: ₹3.4 lakh crore cost overrun, 1,775 projects —



June 2026: ₹4.92 lakh crore overrun, 1,847 projects —



Cost overrun ML on Indian data:



Explainable AI & AI governance in government:



CAG Reports: <https://cag.gov.in>

AI4Bharat / Indic AI Initiatives:

